Do spectral gradient updates help?

auguste-ish

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1 Introduction

Let $d \in \mathbb{N}^*$ and $F : \mathbb{R}^d \to \mathbb{R}$ be a differentiable function.

The gradient descent (GD) method defines sequences $(x_k)_{k\in\mathbb{N}}, (\alpha_k)_{k\in\mathbb{N}}$, such that

$\forall k \in \mathbb{N};\ x_k \in \mathbb{R}^d$, $\alpha_k \in \mathbb{R}_+^*$ and

$$\forall\, k \in \mathbb{N}, \quad x_{k+1} = x_k - \alpha_k \nabla F(x_k)$$

This method is widely used for optimizing objective functions in machine learning. Several variants have been proposed:

- GD with constant steps or decreasing steps
- GD with inertia and/or acceleration: Heavy Ball (HB), Nesterov Acceleration (NAG)
- Stochastic GD (SGD)
- Adaptive and normalized GD methods: AdaGrad, RMSprop, Adam, Sign GD, Spectral GD, Muon

In this study, we focus on the spectral GD method.

Let $(d, p) \in \mathbb{N}^* \times \mathbb{N}^*$ and a function $F : \mathcal{M}_{d,p}(\mathbb{R}) \to \mathbb{R}$, Spectral GD defines the sequences $(x_k)_{k\in\mathbb{N}}, (\alpha_k)_{k\in\mathbb{N}}$, such that $\forall k \in \mathbb{N};\ x_k \in \mathcal{M}_{d,p}(\mathbb{R})$, $\alpha_k \in \mathbb{R}_+^*$ and

$$\forall\, k \in \mathbb{N}, \quad x_{k+1} = x_k - \alpha_k U_k V_k^T$$

Where $\nabla F(x_k) = U_k \Sigma_k V_k^T$ is the thin SVD decomposition of $\nabla F(x_k)$

The aim of this project is to determine, on a few examples, when spectral GD improves the convergence rate of optimization problems compared to other gradient descent methods.

2 MSE loss for linear regression

2.1 Objective function

Definition

Let

- $n, d, p \in \mathbb{N}^*$, $X \in \mathcal{M}_{n,d}(\mathbb{R})$ features, $Y \in \mathcal{M}_{n,p}(\mathbb{R})$ targets
- $\forall \beta = (\beta_{i,j})_{(i,j)\in[\![1,d]\!]\times[\![1,p]\!]} \in \mathcal{M}_{d,p}(\mathbb{R})$, $\|\beta\|_F^2 = \sum_{i,j} \beta_{i,j}^2 = \mathrm{Tr}(\beta^\top \beta)$ (the Frobenius norm)
- $\forall A, B \in \mathcal{M}_{d,p}(\mathbb{R}), \quad \langle A, B \rangle_F = \mathrm{Tr}(A^\top B)$ the Frobenius inner product

Let us consider the Mean Squared Error (MSE) loss for linear regression such that $\forall \beta \in \mathcal{M}_{d,p}(\mathbb{R})$

$$F(\beta) = \frac{1}{np}\|X\beta - Y\|_F^2 \quad , \quad \nabla F(\beta) = \frac{2}{np} X^\top (X\beta - Y) \quad \text{and} \quad \nabla^2 F(\beta) = \frac{2}{np} X^\top X$$

Since $\forall \beta \in \mathcal{M}_{d,p}(\mathbb{R}), \nabla^2 F(\beta) \succeq 0$, F is convex therefore a minimizer β^* of F is such that

$$\nabla F(\beta^*) = \frac{2}{np} X^\top (X\beta^* - Y) = 0 \quad \iff \quad X^\top X \beta^* = X^\top Y$$

Exact solution

X has full column rank $\iff$ $X^\top X$ has full rank

In this case $X^\top X$ is invertible and β^* is such that $\beta^* = (X^\top X)^{-1} X^\top Y$

Therefore $X\beta^* = X(X^\top X)^{-1} X^\top Y := PY$ With $P = X(X^\top X)^{-1} X^\top; \quad P^2 = P; \quad P^\top = P$

The orthogonal projection matrix on span of the columns of X

Approximation of solution by gradient descent

When X does not have full rank, X has full rank but a bad condition number, $X^\top X$ is too large to invert easily ($d \gg 1$, *complexity* $\mathcal{O}(d^3)$ *or using Cholesky* $\mathcal{O}(\frac{d^3}{3})$). In some cases regularized methods (Lasso, Ridge, Elastic Net) give the minimizer of a penalized objective, which can be preferable to OLS. But in all these cases an approximation of a minimizer β^* obtained by GD can still be very satisfying and requires no inversion of $X^\top X$. However, when the conditioning is bad, convergence can be quite slow.

One can notice that:

- $\forall \beta \in \mathcal{M}_{d,p}(\mathbb{R}), \ \nabla^2 F(\beta) \succeq 0$
- F is L-smooth with $L = \frac{2\lambda_{\max}(X^\top X)}{np}$
- If $\lambda_{\min}(X^\top X) > 0$, F is μ-strongly convex with $\mu = \frac{2\lambda_{\min}(X^\top X)}{np}$
- $\mu > 0$ so $\kappa = \frac{L}{\mu} = \frac{\lambda_{\max}(X^\top X)}{\lambda_{\min}(X^\top X)}$

2.2 Spectral GD

Steepest descent optimization

The classic GD sequence (the steepest descent sequence) $x_{k+1} = x_k - \alpha_k \nabla F(x_k)$ is equivalent to

$$x_{k+1} = \underset{x \in \mathcal{M}_{d,p}(\mathbb{R})}{\arg\min} \left\{ F(x_k) + \langle \nabla F(x_k),\, x - x_k \rangle_F + \frac{1}{2\alpha_k} \|x - x_k\|_F^2 \right\}$$

Spectral norm constraint descent optimization

Let

- $(\alpha_k)_{k \in \mathbb{N}}$ be a sequence such that $\forall k, \alpha_k \in \mathbb{R}_+^*$
- $\|A\|_S = \max\left\{\sqrt{\lambda},\ \lambda \in \mathrm{Spec}(A^\top A)\right\}$ be the spectral norm
- $\forall k \in \mathbb{N},\ \alpha_k \mathbb{B}_S = \{x \in \mathcal{M}_{d,p}(\mathbb{R}) \mid \|x\|_S \leq \alpha_k\}$ the ball associated to the spectral norm
- $\iota_{\alpha_k \mathbb{B}_S}(x) = \begin{cases} 0 & \text{if } \|x\|_S \leq \alpha_k \\ +\infty & \text{else} \end{cases}$ be an indicator function

The spectral GD sequence is defined from the ball constrained optimization problem

$$x_{k+1} \in \underset{x \in \mathcal{M}_{d,p}(\mathbb{R})}{\arg\min} \left\{ F(x_k) + \langle \nabla F(x_k),\, x - x_k \rangle_F + \iota_{\alpha_k \mathbb{B}_S}(x - x_k) \right\}$$

which is equivalent to $x_{k+1} = x_k - \alpha_k P_k$ and $P_k = \underset{u \in \mathbb{B}_S}{\arg\max} \langle \nabla F(x_k),\, u \rangle_F = U_k V_k^\top$

Where $\nabla F(x_k) = U_k \Sigma_k V_k^\top$ is the thin SVD of $\nabla F(x_k)$ and P_k the orthogonal polar factor of $\nabla F(x_k)$

Remark

Using the Frobenius norm, $\underset{u \in \mathbb{B}_F}{\arg\max} \langle \nabla F(x_k),\, u \rangle_F = \frac{1}{\|\nabla F(x_k)\|_F} \nabla F(x_k)$

Therefore the Frobenius GD sequence is such that $\forall k \in \mathbb{N},\ x_{k+1} = x_k - \frac{\alpha_k}{\|\nabla F(x_k)\|_F} \nabla F(x_k)$

2.3 GD Methods

2.3.1 Classic and accelerated updates

Notation

- Given L the smoothness constant and $\mu > 0$ the strong convexity constant, so $\kappa = L/\mu$ is finite.
- x_0 is the initial iterate; the momentum methods start with $x_1 = x_0 - \alpha \nabla F(x_0)$.
- The momentum coefficient is written γ here (constant γ^* for HB, iterate-dependent γ_k for NAG). The detailed subsections denote it β and β_k respectively.

For an L-smooth, μ-strongly convex function with $\mu > 0$,

Method	Update rule	Optimal parameters	Asymptotic rate
Classic GD with constant steps	$x_{k+1} = x_k - \alpha \nabla F(x_k)$	$\alpha^* = \frac{2}{\mu + L}$	$\mathcal{O}\left(\left(\frac{\kappa-1}{\kappa+1}\right)^{2k}\right)$
Heavy Ball	$y_k = x_k + \gamma_k(x_k - x_{k-1})$, $x_{k+1} = y_k - \alpha \nabla F(x_k)$	$\alpha^* = \frac{4}{(\sqrt{L}+\sqrt{\mu})^2}$, $\gamma^* = \left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^2$	$\mathcal{O}\left(\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^{2k}\right)$
NAG	$y_k = x_k + \gamma_k(x_k - x_{k-1})$, $x_{k+1} = y_k - \alpha \nabla F(y_k)$	$\alpha^* = \frac{1}{L}$, $\gamma_k = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}$	$\mathcal{O}\left(\left(1 - \frac{1}{\sqrt{\kappa}}\right)^{k}\right)$

Complexity (per iteration): $\mathcal{O}(dp)$

It excludes the gradient evaluation, which costs $\mathcal{O}(ndp)$ for the MSE loss and is common to all methods.

Minimum iterations to reach a tolerance

For a given $\epsilon > 0$, the smallest number of iterations k such that $F(x_k) - F^* \leq \epsilon$, with $R = \|x_0 - x^*\|^2$:

Method	Bound on $F(x_k) - F^*$	$k_{\min}$
Classic GD	$\left(\frac{\kappa-1}{\kappa+1}\right)^{2k} \frac{LR}{2}$	$\left\lceil \frac{\log\left(\frac{2\epsilon}{LR}\right)}{2\log\left(\frac{\kappa-1}{\kappa+1}\right)} \right\rceil$
Heavy Ball	$C_\delta \left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1} + \delta\right)^{2k} R$	$\left\lceil \frac{\log\left(\frac{\epsilon}{C_\delta R}\right)}{2\log\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}+\delta\right)} \right\rceil$
NAG	$\frac{L+\mu}{2}\left(1 - \frac{1}{\sqrt{\kappa}}\right)^{k} R$	$\left\lceil \frac{\log\left(\frac{2\epsilon}{(L+\mu)R}\right)}{\log\left(1-\frac{1}{\sqrt{\kappa}}\right)} \right\rceil$

For HB, $\delta > 0$ is arbitrary and $C_\delta > 0$ is the associated constant; the bound is $(\cdot + \delta)$-suboptimal and the asymptotic rate is recovered as $\delta \to 0$

2.3.2 Norm-constrained gradient steps

$H_k := \nabla^2 F(x_k)$, $G_k := \nabla F(x_k)$

Method	Update rule	Step	Asymptotic rate	Complexity (per iteration)
Frobenius GD	$x_{k+1} = x_k - \frac{\alpha_k}{\|G_k\|_F} G_k$	$\alpha_k = \frac{\|G_k\|_F^3}{\langle H_k G_k, G_k \rangle_F}$	$\mathcal{O}\left(\left(\frac{\kappa-1}{\kappa+1}\right)^{2k}\right)$	$\mathcal{O}(d^2 p)$
Spectral GD	$x_{k+1} = x_k - \alpha_k P_k$	$\alpha_k = \frac{\langle G_k, P_k \rangle_F}{\langle H_k P_k, P_k \rangle_F}$	$\mathcal{O}\left(\left(\frac{\kappa-1}{\kappa+1}\right)^{2k}\right)$	$\mathcal{O}(dp \min(d,p) + d^2 p)$

Complexity with the Hessian precomputed once

Method	Bound on $F(x_k) - F^*$	$k_{\min}$
Frobenius GD	$\left(\frac{\kappa-1}{\kappa+1}\right)^{2k} \frac{LR}{2}$	$\left\lceil \frac{\log(\frac{2\epsilon}{LR})}{2\log(\frac{\kappa-1}{\kappa+1})} \right\rceil$
Spectral GD	$\left(\frac{\kappa-1}{\kappa+1}\right)^{2k} \frac{LR}{2}$	$\left\lceil \frac{\log(\frac{2\epsilon}{LR})}{2\log(\frac{\kappa-1}{\kappa+1})} \right\rceil$

This is a worst-case bound; in practice stop when $\|\nabla F(x_k)\| \leq \text{tol}$ or $F(x_k) - F^* \leq \epsilon$

2.4 Theoric conclusions

For this objective function, theoretically speaking,

- All methods have an asymptotic rate that depend on the conditioning
- The Cauchy steps Spectral GD sequence does not have a better asymptotic rate than classic, inertia and accelerated methods
- The Cauchy steps Spectral GD sequence is more costly to compute

2.5 Application on a dataset

Tsanas, A. & Xifara, A. (2012). Energy Efficiency Dataset. UCI Machine Learning Repository. https://doi.org/10.24432/C51307.

It contains $n = 768$ samples described by $d = 8$ features and $p = 2$ continuous targets. Since we are interested in the optimization dynamics rather than in predictive performance, we do not split the data into training and test sets.

	count	mean	std	min	25%	50%	75%	max
X1	768.0	0.764167	0.105777	0.62	0.6825	0.75	0.8300	0.98
X2	768.0	671.708333	88.086116	514.50	606.3750	673.75	741.1250	808.50
X3	768.0	318.500000	43.626481	245.00	294.0000	318.50	343.0000	416.50
X4	768.0	176.604167	45.165950	110.25	140.8750	183.75	220.5000	220.50
X5	768.0	5.250000	1.751140	3.50	3.5000	5.25	7.0000	7.00
X6	768.0	3.500000	1.118763	2.00	2.7500	3.50	4.2500	5.00
X7	768.0	0.234375	0.133221	0.00	0.1000	0.25	0.4000	0.40
X8	768.0	2.812500	1.550960	0.00	1.7500	3.00	4.0000	5.00
Y1	768.0	22.307201	10.090196	6.01	12.9925	18.95	31.6675	43.10
Y2	768.0	24.587760	9.513306	10.90	15.6200	22.08	33.1325	48.03

Figure 1: Describe features

We center and reduce (z-score) all features

2.5.1 Features Subsets

Features linear dependence, correlation and conditioning

	features_subset	features	count_features	rank_X	is_X_full_column_rank	rank_G	is_G_full_rank	cond_nb_G	L	mu	kappa
0	1	X1|X3|X6|X7|X8	5	5	True	5	True	1.54	1.211384	0.786010	1.54
1	2	X1|X3|X4|X6|X7|X8	6	6	True	6	True	268.05	1.871265	0.006981	268.05
2	3	X1|X2|X3|X4|X5|X6|X7|X8	8	7	False	7	False	40,300,000.00	3.698119	0.000000	35,900,000.00

Figure 2: Features subsets

	X1	X3	X6	X7	X8
X1	1.000000	0.203800	0.000000	0.000000	0.000000
X3	0.203800	1.000000	0.000000	0.000000	0.000000
X6	0.000000	0.000000	1.000000	0.000000	0.000000
X7	0.000000	0.000000	0.000000	1.000000	0.213000
X8	0.000000	0.000000	0.000000	0.213000	1.000000

(a) Subset 1

	X1	X3	X4	X6	X7	X8
X1	1.000000	0.203800	0.868800	0.000000	0.000000	0.000000
X3	0.203800	1.000000	0.292300	0.000000	0.000000	0.000000
X4	0.868800	0.292300	1.000000	0.000000	0.000000	0.000000
X6	0.000000	0.000000	0.000000	1.000000	0.000000	0.000000
X7	0.000000	0.000000	0.000000	0.000000	1.000000	0.213000
X8	0.000000	0.000000	0.000000	0.000000	0.213000	1.000000

(b) Subset 2

	X1	X2	X3	X4	X5	X6	X7	X8
X1	1.000000	0.991900	0.203800	0.868800	0.827700	0.000000	0.000000	0.000000
X2	0.991900	1.000000	0.195500	0.880700	0.858100	0.000000	0.000000	0.000000
X3	0.203800	0.195500	1.000000	0.292300	0.281000	0.000000	0.000000	0.000000
X4	0.868800	0.880700	0.292300	1.000000	0.972500	0.000000	0.000000	0.000000
X5	0.827700	0.858100	0.281000	0.972500	1.000000	0.000000	0.000000	0.000000
X6	0.000000	0.000000	0.000000	0.000000	0.000000	1.000000	0.000000	0.000000
X7	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	1.000000	0.213000
X8	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.213000	1.000000

(c) Subset 3

Figure 3: Features correlations

- Subset 1: In this case, there are no linear dependence no correlation, $\kappa = 1.54$
- Subset 2: In this case, X is full column rank, the features are not linearly dependent. However because of some features being highly correlated (X_1, X_4) the conditioning is moderately bad $\kappa = 268.05$
- Subset 3 (All features): For this subset one can see that X is not full column rank. The columns of X are linearly dependent hence $X^\top X$ is not invertible.

2.5.2 Results

2.5.2.1 Figures

	features_subset	method	kappa	lr	mom	delta	asymptotic_rate	epsilon	epochs
0	1	classicGD	1.541181	1.001305	NaN	NaN	0.212964	0.00001	4
1	1	HB	1.541181	1.012923	0.011603	NaN	0.107718	0.00001	3
2	1	NAG	1.541181	0.825502	0.107718	0.00001	0.194486	0.00001	8
3	1	cauchyFrobeniusGD	1.541181	Not fixed	NaN	NaN	0.212964	0.00001	4
4	1	cauchySpectralGD	1.541181	Not fixed	NaN	NaN	0.212964	0.00001	4

(a) Subset 1 optimizers params

	features_subset	method	kappa	lr	mom	delta	asymptotic_rate	epsilon	epochs
0	2	classicGD	268.053314	1.064823	NaN	NaN	0.992567	0.00001	768
1	2	HB	268.053314	1.898583	0.783003	NaN	0.884874	0.00001	48
2	2	NAG	268.053314	0.534398	0.884874	0.00001	0.938921	0.00001	182
3	2	cauchyFrobeniusGD	268.053314	Not fixed	NaN	NaN	0.992567	0.00001	768
4	2	cauchySpectralGD	268.053314	Not fixed	NaN	NaN	0.992567	0.00001	768

(b) Subset 2 optimizers params

Figure 4: Optimizers parameters for both subsets

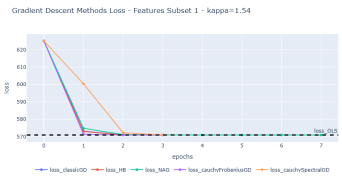
	features_subset	OLS	GD	HB	NAG	CauchyFrobeniusGD	CauchySpectralGD
0	1	570.946472	570.950317	570.990784	570.946472	570.946594	571.004272
1	2	562.169067	562.169739	567.010559	562.169067	562.169067	562.169067

(a) Loss function last values

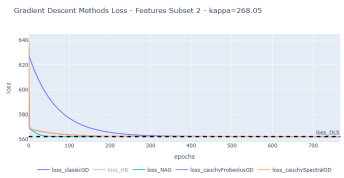
	features_subset	GD	HB	NAG	CauchyFrobeniusGD	CauchySpectralGD
0	1	0.0000	0.0002	0.0	0.000	0.0006
1	2	0.0017	0.4037	0.0	0.001	0.0017

(b) Distance to OLS weights

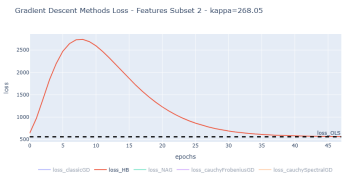
Figure 5: Final loss values and distance to OLS weights



(a) Subset 1 loss



(b) Subset 2 loss (without HB)



(c) Subset 2 loss (with HB)

Figure 6: Loss curves across methods

2.5.2.2 Comments

For subset 1 ($\kappa = 1.54$) the problem is well conditioned and all methods reach the OLS loss within a few epochs; from epoch 2 onwards the loss curves are almost indistinguishable.

For subset 2 ($\kappa = 268.05$) classical GD is the slowest and needs hundreds of epochs to approach the OLS loss. In contrast, the Cauchy spectral step reaches the OLS loss almost immediately, while Heavy Ball strongly overshoots (the loss grows to about 2700) before converging.

The results are aligned with the theoretical asymptotic rates given the conditionings. Overall, spectral updates clearly help on the ill-conditioned subset, while the well-conditioned subset does not distinguish the methods.

3 Todo

- Why Cauchy Steps Spectral GD better start than classic GD ??
- Test other objective functions
- Test other GD methods, other steps for Spectral GD

4 References

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